go much further, being trained on exponentially larger datasets.

- ***Architecture:*** This refers to the underlying structure of the model. For instance, GPT-3.5 utilises a transformer architecture, which is particularly adept at handling the complexities of natural language due to its ability to process sequences of data and capture contextual relationships within text.
- ***Training method:*** The process involves iterative learning, where the model, using its architecture, learns to predict and generate text based on the input data it's fed. This training is computationally intensive, requiring significant processing power and, consequently, financial resources.

Building and even fine-tuning LLMs demand considerable computational resources, making them costly ventures. Recognising this, certain organisations have stepped forward to bear the financial burden, training these sophisticated models and then releasing them to the open source community. Their contributions ensure that the benefits of LLMs can be accessed and further developed by researchers, developers, and innovators worldwide, fostering a collaborative ecosystem of AI advancement.

The philosophy behind open source LLMs

The essence of open source LLMs lies in the belief that technological advancements should be a collaborative, shared resource. Organisations like Meta have pioneered this movement, training and releasing state-of-the-art models like LLaMA 2 and the upcoming LLaMA 3. These models, while free to use, are built upon the significant financial investments of their parent companies, reflecting a commitment to open source principles that transcend traditional business models.

Two examples of pioneering open source LLMs are:

- ***LLaMA series by Meta:*** A testament to Meta's investment in the open source community, these models serve as robust platforms for further innovation. They enable creators to develop solutions that are as diverse as the use cases they envision—from enhancing conversational AI to creating new forms of digital interaction.
- ***Falcon models by the Technology Innovation Institute:*** The Falcon models embody the practical application of open source philosophy, offering efficiency and a wide knowledge base, thanks to their training on extensive datasets like RefinedWeb. They exemplify how open source can bridge the gap between cutting-edge research and real-world application.

In this context, 'open source' signifies more than code; it's a commitment to collective progress in AI. It represents a future where the most advanced tools in natural language processing are not the exclusive domain of the few but a shared asset for the many, driving forward the boundaries of what we can achieve with technology.

Potential risks of ignoring open source LLMs

Over-reliance on proprietary models

The convenience of proprietary LLMs comes with a hidden cost: over-reliance. This dependency is not just about the technological aspects but extends to decision-making processes where the opacity of these models can lead to unquestioned acceptance of

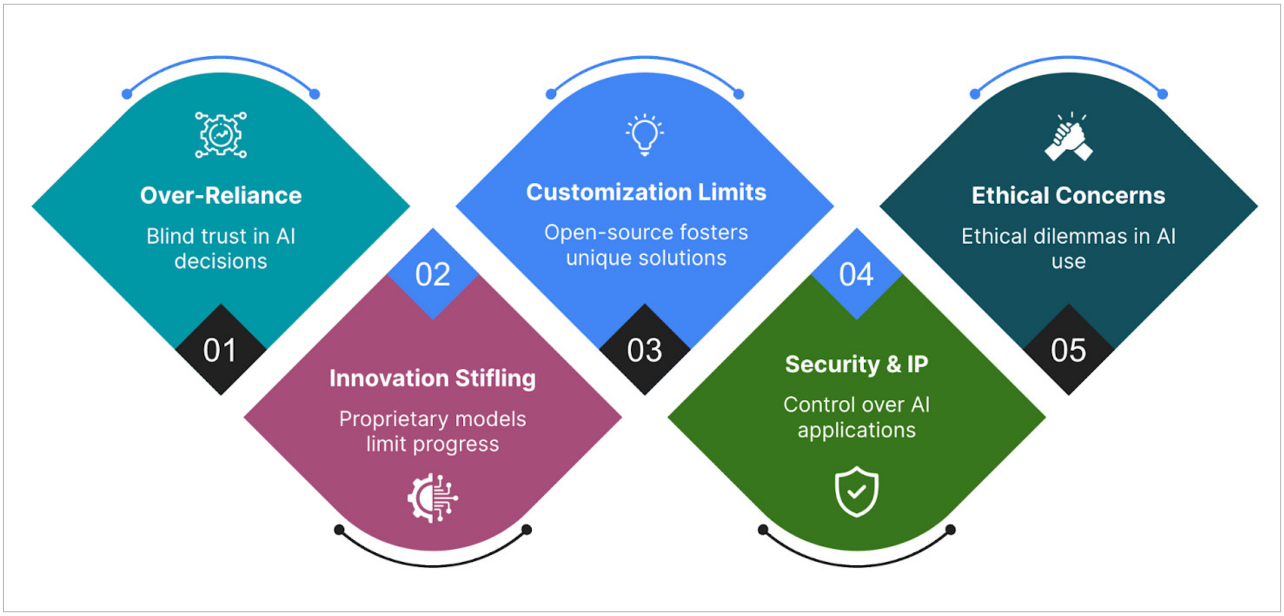

Figure 1: Pictorial representation of potential risks of ignoring open source LLMs